MSc Machine Learning & Data Science
Exploratory Data Analysis and Visualisation
Coursework 2 – Automated Karyotyping

HAND OUT: 9:00, Thursday 12 March 2026
HAND IN: 9:00, Wednesday 1 April 2026

Weight: 50% of module
Expected effort: 12 hours
Individual work

Overview

In this coursework you will demonstrate your ability to carry out **exploratory data analysis (EDA)** of multivariate data, reason about the impact of **missingness and preprocessing choices** on downstream exploratory conclusions, and communicate findings clearly for a non-specialist audience.

The coursework is organised into **three questions** that reflect a realistic workflow:

- initial exploration, data quality assessment, and variable selection;
- visual methods to implicate cluster structure, including a comparison of two alternative pipelines;
- summarisation of cluster structure using prototypes and low-dimensional representations.

This assignment must be attempted individually; your submission must be your own, unaided work. Candidates are prohibited from discussing assessed coursework, and must abide by Imperial College's rules regarding academic integrity and plagiarism. The submission of output from generative AI tools (e.g., ChatGPT) for this coursework is prohibited. Violations will be treated as an examination offence. Enabling other candidates to plagiarise your work constitutes an examination offence. To ensure quality assurance is maintained, departments may choose to invite a random selection of students to an 'authenticity interview' on their submitted assessments.

Framing

In automated karyotyping, we seek to identify the *chromosome class* of each chromosome from features extracted from cell images (karyograms). These quantitative features are produced by image analysis and capture aspects of chromosome **size and shape** as well as a chemically induced **banding (staining) pattern**.

In this coursework you are given a dataset of chromosome feature vectors. The **true chromosome class labels are not provided**. Your task is not to recover the true labels, but to use EDA and visual clustering methods to infer **how many distinct chromosome classes are present in the dataset**.

The dataset contains missing values. It may also contain variables that are uninformative artefacts of feature extraction.

You may assume that there are **more than 1 and fewer than 8** underlying groups present.

There is no single "correct" answer. Marks are awarded for reasoning, evidence, clarity, and justification.

Dataset

The file `chromosome_data.csv`, provided on Blackboard, contains:

- 832 observations,
- 33 numerical variables,
- missing values distributed across the dataset.

All variables are numeric and on their original reported scales.

Sampling

You must:

1. Load the dataset.
2. Set the random seed equal to the **last three digits** of your College ID.
3. Sample 600 rows without replacement.
4. Conduct all analysis using this sampled dataset only.

Failure to follow this sampling protocol may affect marks.

Question 1 – Exploratory Data Analysis (30%)

The aim of Q1 is to understand the structure of the dataset and make defensible preprocessing decisions. You should:

- Report the dimensions of the data and summarise variable scales.
- Quantify and visualise missingness (by variable and/or observation).
- Examine structure using appropriate multivariate tools seen in the labs (e.g. correlation analysis, PCA).
- Identify any variables that appear uninformative or anomalous, and justify whether you retain or remove them.
- Discuss whether scaling is appropriate for subsequent clustering and justify your decision.

This section should demonstrate structured EDA, not a sequence of unconnected plots.

Question 2 – Cluster Structure and Comparison (40%)

You will compare two clustering pipelines. In each case, consider

$$k \in \{2, 3, 4, 5, 6, 7\}$$

and use elbow plots to guide your choice of k.

Standard Euclidean Route

- Choose a missing-data handling strategy: casewise deletion or column-mean imputation.
- Justify your choice briefly.
- Run k-means for $k = 2, \ldots, 7$.
- Present elbow plot(s) and justify your selected k.
- Provide appropriate visualisation of the resulting clusters (e.g. PCA scores coloured by cluster).

Observed-Only Distance Route

- Compute the renormalised Euclidean distance using observed values only:

$$d_{ij}^{(\mathrm{obs})} = \sqrt{\frac{p}{|O_{ij}|} \sum_{k \in O_{ij}} (x_{ik} - x_{jk})^2}$$

where O_{ij} is the set of coordinates observed for both observations i and j.

- Embed the resulting distance matrix using a method covered in the labs (e.g. MDS).
- Run k-means on the embedded coordinates.
- Present elbow plot(s) and justify your selected k.
- Provide appropriate visualisation of the clusters in the embedded space.

Comparison

- Report the proportion of differing assignments between the two approaches.
- Comment on whether the two clusterings are broadly similar or meaningfully different.
- Discuss possible reasons for differences (e.g. missing-data handling, scaling, geometry).

Question 3 – Cluster Interpretation (30%)

Using your preferred clustering solution:

- Construct cluster prototypes (e.g. centroids).
- Use appropriate visualisations (e.g. parallel coordinates, PCA, boxplots) to describe how clusters differ.
- State clearly how many chromosome classes you believe are present.
- Discuss within-cluster variability.
- Acknowledge uncertainty and sensitivity to modelling choices.

Your interpretation should be structured and evidence-based.

Deliverables

You must submit:

- **R Markdown** (fully reproducible; code and commentary clearly structured).
- **Report** (maximum 5 pages).
- **Video presentation** (maximum 2 minutes), aimed at an educated but non-expert audience.

Assessment Criteria

Marks are based on:

- Quality of reasoning and justification.
- Appropriate use of methods covered in the labs.
- Clarity and structure of analysis.
- Evidence supporting conclusions.
- Recognition of uncertainty and methodological sensitivity.

Use of methods not covered in the labs is not required and will not attract additional credit.

Academic Integrity

This is an individual piece of work. The usual policies on plagiarism, collaboration, and use of generative AI apply. You may be asked to discuss or explain your submitted work.